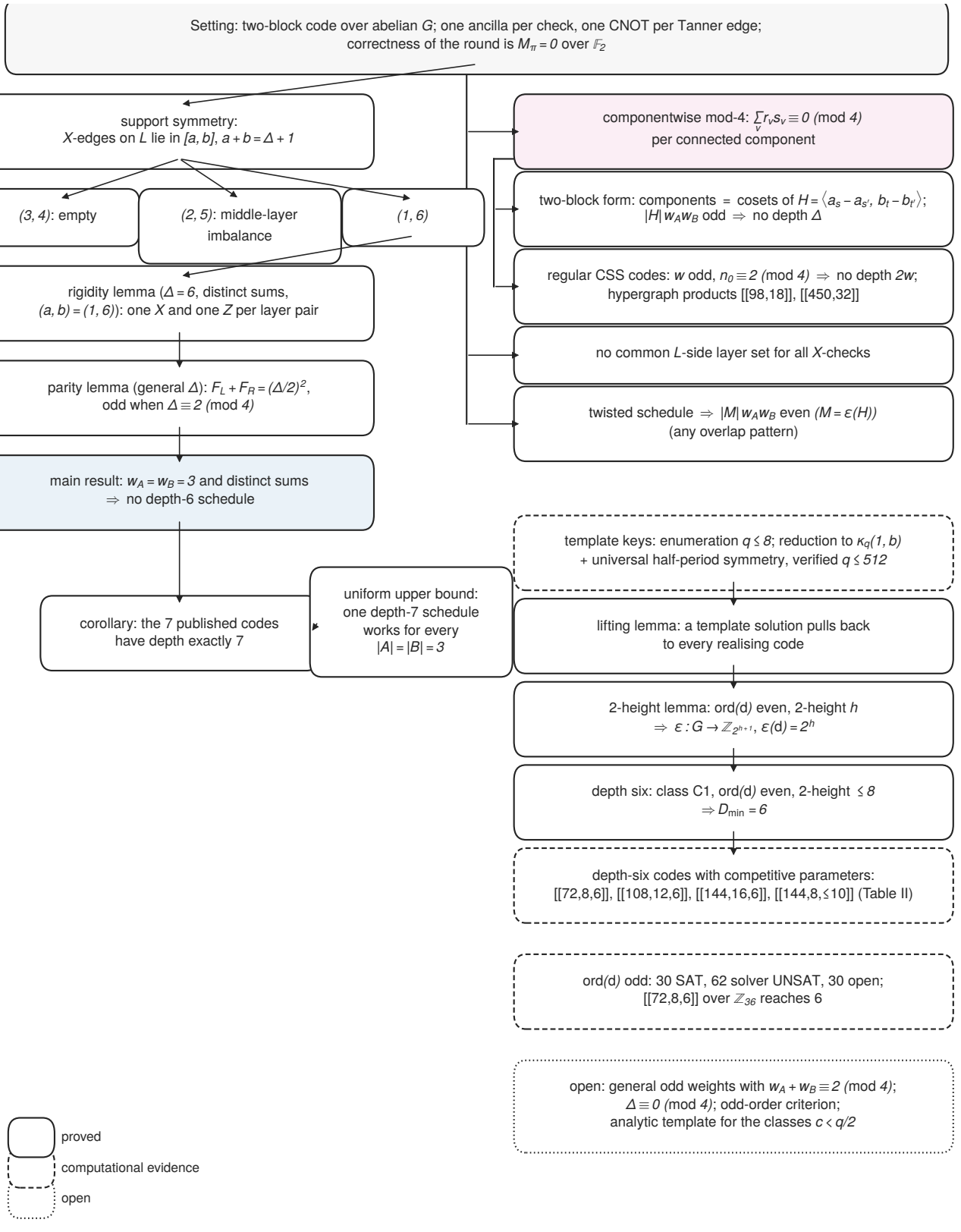
Setting: two-block code over abelian $G$; one ancilla per check, one CNOT per Tanner edge; correctness of the round is $M_\pi = 0$ over $\mathbb{F}_2$
support symmetry: $X$-edges on $L$ lie in $[a, b]$, $a + b = \Delta + 1$
$(3, 4)$: empty
$(2, 5)$: middle-layer imbalance
$(1, 6)$
rigidity lemma ($\Delta = 6$, distinct sums, $(a, b) = (1, 6)$): one $X$ and one $Z$ per layer pair
parity lemma (general $\Delta$): $F_L + F_R = (\Delta/2)^2$, odd when $\Delta \equiv 2 \pmod 4$
main result: $w_A = w_B = 3$ and distinct sums $\Rightarrow$ no depth-6 schedule
corollary: the 7 published codes have depth exactly 7
uniform upper bound: one depth-7 schedule works for every $|A| = |B| = 3$
componentwise mod-4: $\sum_v r_v s_v \equiv 0 \pmod 4$ per connected component
two-block form: components = cosets of $H = \langle a_s - a_{s'},\ b_t - b_{t'} \rangle$; $|H|\, w_A w_B$ odd $\Rightarrow$ no depth $\Delta$
regular CSS codes: $w$ odd, $n_0 \equiv 2 \pmod 4$ $\Rightarrow$ no depth $2w$; hypergraph products [[98,18]], [[450,32]]
no common $L$-side layer set for all $X$-checks
twisted schedule $\Rightarrow$ $|M|\, w_A w_B$ even $(M = \varepsilon(H))$ (any overlap pattern)
template keys: enumeration $q \le 8$; reduction to $\kappa_q(1, b)$ + universal half-period symmetry, verified $q \le 512$
lifting lemma: a template solution pulls back to every realising code
2-height lemma: ord(d) even, 2-height $h$ $\Rightarrow$ $\varepsilon : G \to \mathbb{Z}_{2^{h+1}}$, $\varepsilon(\mathrm{d}) = 2^h$
depth six: class C1, ord(d) even, 2-height $\le 8$ $\Rightarrow D_{\min} = 6$
depth-six codes with competitive parameters: [[72,8,6]], [[108,12,6]], [[144,16,6]], [[144,8,≤10]] (Table II)
ord(d) odd: 30 SAT, 62 solver UNSAT, 30 open; [[72,8,6]] over $\mathbb{Z}_{36}$ reaches 6
open: general odd weights with $w_A + w_B \equiv 2 \pmod 4$; $\Delta \equiv 0 \pmod 4$; odd-order criterion; analytic template for the classes $c < q/2$
proved
computational evidence
open
published codes: DRAT, drat-trim verified, no symmetry reduction